

# Multiplicity Operator Calculus (MOC)

Here's a compact "Multiplicity Operator Calculus" (MOC) with notation, sample operators for  $p \in \{2, 3, 5, 7\}$ , and a worked 108-cycle generator.

## 1) Primitives and notation

- **Time lattice:**  $\mathbb{T}_n = \mathbb{Z}/n\mathbb{Z}$  with ticks  $t = 0, \dots, n-1$ .
- **Pattern:**  $x : \mathbb{T}_n \rightarrow \mathbb{R}^k$  (e.g., amplitude, accent, color).
- **Operator word:**  $\mathcal{W} = \hat{O}_m \cdots \hat{O}_2 \hat{O}_1$ . Apply right-to-left:  $x \mapsto \mathcal{W}[x]$ .
- **Prime family:** for each prime  $p$  and power  $p^r$ , define  $\hat{P}_p^{(r)} = \hat{P}_p^r$ .
- **Divisibility indicator:**  $[d \mid t] = 1$  if  $t \equiv 0 \pmod{d}$ , else 0.

## 2) Core operator types

Each prime contributes one canonical operator in four "modes" that act on  $x$  over  $\mathbb{T}_n$ . These modes commute within a prime but not across primes, in general.

1) **Phase-gate (accent)**  $A_{p^r}$  : adds weighted accents at period  $p^r$ .

$$(A_{p^r}x)(t) = x(t) + \alpha_{p^r} [p^r \mid t] e_1$$

$e_1$  picks the accent channel;  $\alpha_{p^r} \in \mathbb{R}_{\geq 0}$ .

2) **Subdivision (refine)**  $S_p$  : lifts  $\mathbb{T}_n \rightarrow \mathbb{T}_{pn}$  by  $t \mapsto pt + u$ ,  $u \in \{0, \dots, p-1\}$ .

$$(S_p x)(pt + u) = x(t) \quad \text{for all } u$$

3) **Rotation (cyclic shift)**  $R_{p^r}^\phi$  : shifts by  $\phi$  on the  $p^r$ -grid.

$$(R_{p^r}^\phi x)(t) = x(t + \phi [t]_{p^r})$$

where  $[t]_{p^r}$  projects  $t$  to its residue class in  $\mathbb{Z}/p^r\mathbb{Z}$ .

4) **Permutation (wreath)**  $W_p$  : reorders the  $p$  children created by  $S_p$ . For a fixed permutation  $\pi \in S_p$ :

$$(W_p x)(pt + u) = x(pt + \pi(u))$$

## 3) Prime-indexed sample operators

Choose clear semantics per prime. These are defaults; swap as needed.

### $p = 2$ (binary microstructure)

- $S_2$  : double resolution.
- $A_{2^r}$  : downbeat at multiples of  $2^r$  .
- $R_{2^r}^1$  : flip even/odd at level  $2^r$  .
- $W_2$  with  $\pi = (0\ 1)$  : swap the two subticks  $\rightarrow$  off-beat emphasis.

### $p = 3$ (ternary phrasing)

- $S_3$  : triple subdivision.
- $A_{3^r}$  : cadence at 3, 9, 27, ...
- $R_{3^r}^1$  : rotate within each  $3^r$  cell.
- $W_3$  with  $\pi = (0\ 2\ 1)$  : hemiola-like reorder.

### $p = 5$ (quinary color / ornament)

- $S_5$  : quintuple filigree.
- $A_5$  : light ornament every 5th tick.
- $R_5^\phi$  : micro-swing on a 5-grid.
- $W_5$  with  $\pi = (0\ 2\ 4\ 1\ 3)$  : pentachord rotation.

### $p = 7$ (heptadic asymmetry)

- $S_7$  : sevenfold lace.
- $A_7$  : marker every 7th tick.
- $R_7^\phi$  : skew within septets.
- $W_7$  with a single 7-cycle: long rotating accent.

## 4) Noncommutation and calculus rules

Let primes  $p / q$  **Same-prime modes commute** up to a predictable phase:

$$A_{p^a} R_{p^b}^\phi = R_{p^b}^\phi A_{p^a} .$$

$S_p$  commutes with  $A_{p^r}$  after index lift. - **Cross-prime braid**:

$$S_p A_{q^r} = A_{q^r}^{\uparrow p} S_p, \quad W_p A_{q^r} / A_{q^r} W_p \text{ in general}$$

where  $\uparrow p$  means “copy accents to each of the  $p$  children,” which changes feel. - **Operator word comparison**: order matters. Define

$$\mathcal{W}_1 \prec \mathcal{W}_2 \iff \text{their outputs differ on at least one tick.}$$

## 5) Resonance functional

Given target data  $D : \mathbb{T}_n \rightarrow \mathbb{R}^k$  and model  $x$  ,

$$R(x; D) = 1 - \frac{\|x - D\|}{\|D\|}$$

with any norm meaningful to the task. “Proof by resonance” = maximize  $R$  by editing the operator word.

## 6) Minimal construction sutras

- **S1 (Factor-Fold):** If  $n = \prod p_i^{r_i}$ , create  $\mathbb{T}_n$  by nested  $S_{p_i}$  in any order, then gate accents with  $A_{p_i^{r_i}}$ .
- **S2 (Order-Matters):** Change feel by permuting the operator word; keep the multiset  $\{p_i^{r_i}\}$  fixed.
- **S3 (Local-Global Lock):** Use  $R_{p^r}$  to align micro onsets with macro cadences.

## 7) Worked 108-cycle generator

Goal: build  $n = 108 = 2^2 \cdot 3^3$  with clear ternary phrasing and binary micro-pulse.

### 7.1 Operator word

Pick the “ternary-first” feel:

$$\mathcal{W}_{108} = \underbrace{S_2 S_2}_{\times 4} \underbrace{S_3 S_3 S_3}_{\times 27} \cdot A_{27} A_9 A_3 A_4 A_2$$

Interpretation: - Build  $\mathbb{T}_{27}$  via  $S_3$  three times, then lift to  $\mathbb{T}_{108}$  via  $S_2^2$ .

- Gate accents at levels 27, 9, 3 (ternary) and 4, 2 (binary).

### 7.2 Explicit accent logic on $\mathbb{T}_{108}$

For tick  $t \in \{0, \dots, 107\}$ , define hierarchical flags:

$$a_3(t) = [3 \mid t], \quad a_9(t) = [9 \mid t], \quad a_{27}(t) = [27 \mid t], \quad b_2(t) = [2 \mid t], \quad b_4(t) = [4 \mid t].$$

Composite weight (choose any monotone scheme), e.g.

$$W(t) = 4 a_{27}(t) + 3 a_9(t) + 2 a_3(t) + 1 b_4(t) + \frac{1}{2} b_2(t).$$

Onset rule: - Place a primary hit if  $a_3(t) = 1$ .

- Strengthen if  $a_9(t) = 1$  and again if  $a_{27}(t) = 1$ .

- Fill light micro-ticks where  $b_2$  or  $b_4$  fire but  $a_3 = 0$ .

This yields 36 ternary beats ( $108/3$ ) grouped as 27-grid cadences every 27 ticks, with 9-level sub-cadences.

### 7.3 Two feels via noncommutation

- **Ternary-first** (above):  $S_3^3$  then  $S_2^2$ . Accents  $A_{27}, A_9, A_3$  are copied to all binary children, giving a stable ternary phrase over binary micro-pulse.
- **Binary-first**: swap build order

$$\mathcal{W}'_{108} = S_3^3 S_2^2 \cdot W_2 A_4 A_2 A_{27} A_9 A_3.$$

Now  $W_2$  offsets micro onsets before ternary gating, yielding a syncopated feel.  $\mathcal{W}'_{108} \prec \mathcal{W}_{108}$ .

## 7.4 Canonical tick classes for 108

Class labels by divisibility: -  $C_0 : t \equiv 0 \pmod{27} \rightarrow$  cadence apex.

-  $C_1 : t \equiv 0 \pmod{9}$  but  $\not\equiv 0 \pmod{27} \rightarrow$  strong.

-  $C_2 : t \equiv 0 \pmod{3}$  but  $\not\equiv 0 \pmod{9} \rightarrow$  medium.

-  $C_3 : t \equiv 0 \pmod{4}$  but  $\not\equiv 0 \pmod{3} \rightarrow$  light binary.

-  $C_4 : t \equiv 0 \pmod{2}$  but  $\not\equiv 0 \pmod{4}$  and  $\not\equiv 0 \pmod{3} \rightarrow$  ghost.

- **Else:** silence or sustain.

## 7.5 Optional color layers from $p = 5, 7$

- Add  $A_5$  for periodic ornaments without disturbing the 108 scaffold:

$$\mathcal{W}_{108+5} = \mathcal{W}_{108} \cdot A_5.$$

- Use  $R_7^\phi$  to drift a heptadic line over the 108 grid:

$$\mathcal{W}_{108+7} = \mathcal{W}_{108} \cdot S_7 W_7 R_7^\phi.$$

Because  $7 \nmid 108$ , this creates a slow-moving phase weave.

## 8) One-line sutra for 108

"Make 27, lace it by 4, mark  $27 \rightarrow 9 \rightarrow 3$ , let  $4 \rightarrow 2$  whisper between."

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**Next:** request a concrete tick list or a machine-readable pattern file (CSV, JSON, or MIDI grid) if needed.